Pre-Analysis Plan

Wladislaw Mill* Felix Rusche†

July 21, 2025

Hypothesis I: Having daughters leads to stronger support for gender equality (as measured by attitudes).

Variables Outcome: The outcome variable is an index of gender attitudes surveyed in 2012. This includes four survey items on a four-point scale:

- women should care more about family than their career
- men should participate in the household just as much as women
- men should care more for the family's financial security than women
- it's best if mothers and fathers equally participate in the labor market and the household

(own translation of survey items, see link for the full survey (p. 103f.).)

We rescale all variables such that a value of 4 is associated with greater gender equality. Following this, we add them up and rescale the resulting variable between 0 and 1, 1 meaning that the individual fully supports gender equality across the four survey items.

Treatment: The treatment variable is the number of daughters an individual has had.

Controls: The controls include a second-degree polynomials of age to control for effects across generations. Further, we control for the number of children using a dummy variable to control for mechanic effect of having more children leading to more daughters and for differences between people with more/fewer children. Finally, we control for gender using a dummy variable for the respondent being male.

*University of Mannheim, mill@uni-mannheim.de
†University of Mannheim & Max Planck Institute for Research on Collective Goods, felix.rusche@uni-mannheim.de

Inclusion/Exclusion Criteria The question implies that those with daughters have been treated. Given that the aim is to measure the effect of having a child with a certain sex rather than of having any children or the number thereof, we exclude individuals without any children. Moreover, we only include private households that finished the full survey.

Exact Analysis Conditional on having an additional child, having a daughter is random, given few instances of selective abortions in Germany. Given that the data on outcomes is only available in 2012, we further use a simple OLS regression framework:

$$GenderAttitudes_i = \beta_0 + \beta_1 Daughters_i + \beta_2 Male_i + \beta_3 Age_i + \beta_4 Age_i^2 + \gamma_i + \varepsilon_i$$

where $Daugthers_i$ counts the number of daughters and γ_i is a vector of dummy variables for the number of children individual i has had. Age_i and $Male_i$ describe the control variables discussed above. We use standard errors robust to heteroscedasticity.

Interpretation of Results A positive coefficient of β_1 suggests that having daughters leads to stronger support for gender equality. Aside from the randomness of gender at birth, the effect hinges on the assumption that parents decision on how many children to have does not depend on the gender of their existing children. Moreover, it's important to note that after conditioning on the number of children the count of sons and daughters is linearly dependent. As such, we cannot differentiate the effect of having an additional daughter from having one fewer son.

Hypothesis II: Having daughters leads to more left-leaning political preferences.

Variables The dependent variable is defined as the 0-10 scale of political orientation. Respondents are asked to place themselves on a political scale from 0 (left) to 10 (right).

We use the same controls as defined above. We additionally control for survey year fixed effects to take out variation in political preferences over time. Moreover, we use a fixed effects panel data model (within estimator).

Inclusion/Exclusion Criteria Same as above.

Exact Analysis We run the following Fixed-effects panel regression

$$LeftRight_{i,t} = \beta_0 + \beta_1 Daughters_{i,t} + \beta_2 Male_{i,t} + \beta_3 Age_{i,t} + \beta_4 Age_{i,t}^2 + \gamma_{i,t} + \theta_i + \omega_t + \varepsilon_{i,t}$$

where $Daugthers_{i,t}$ counts the number of daughters and $\gamma_{i,t}$ is a vector of dummy variables for the number of children individual i has had by time t. $Age_{i,t}$ and $Male_{i,t}$ describe the control variables discussed above (note: male is omitted automatically, unless there are any

gender-changes observed in the data). In addition, we include survey year dummies (θ_t) and individual level fixed effects (ω_i). We use standard errors clustered at the individual level.

Interpretation of Results If β_1 is negative, this suggests that having daughters is related to more left-leaning political preferences. The interpretation is subject to the same assumptions as for H1, including the linear dependence with having had fewer sons.

Hypothesis III: Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).

Variables As the dependent variable of interest, we compute the average number of hours women spend on housework per day. As we only include women in the regression, we remove the male dummy. Aside from that, the variables included remain equivalent to those used in hypothesis II.

Inclusion/Exclusion Criteria Aside from the inclusion criteria stated for hypothesis I, we additionally restrict the sample to women only, given that the outcome variable of interest is the number of hours women work in the household.

Exact Analysis We run the following FE panel

$$HHwork_{i,t} = \beta_0 + \beta_1 Daughters_{i,t} + \beta_2 Age_{i,t} + \beta_3 Age_{i,t}^2 + \gamma_i + \theta_i + \omega_t + \varepsilon_{i,t}$$

where $Daugthers_{i,t}$ counts the number of daughters and $\gamma_{i,t}$ is a vector of dummy variables for the number of children individual i has had by time t. $Age_{i,t}$ controls for the respondent's age. In addition, we include survey year dummies (θ_t) and individual level fixed effects (ω_i). We use standard errors clustered at the individual level.

Interpretation of Results A negative coefficient of β_1 would suggest that women work less in the household as a result of having daughters. The interpretation is subject to the same assumptions as for H1, including the linear dependence with having had fewer sons. Moreover, we need to assume that female children do not produce differential amounts of household work and do not participate more in the household themselves. This last assumption may be violated.

Hypothesis IV: Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

Variables We use the same variables as for H1. The only difference is an interaction term between the respondent being male and the number of daughters as the treatment variable of interest.

Inclusion/Exclusion Criteria Same as for H1.

Exact Analysis Same as for H1. The only difference is that we now include an interaction term between being male and having daughters:

$$\begin{aligned} GenderAttitudes_i = &\beta_0 + \beta_1 Daughters_i \times Male_i + \beta_2 Daughters_i + \beta_3 Male_i + \beta_4 Age_i + \\ &\beta_5 Age_i^2 + \gamma_i + \varepsilon_i \end{aligned}$$

Interpretation of Results A positive coefficient β_1 suggests that men's gender attitudes change stronger in the direction of gender equality than women's attitudes when having a daughter. The interpretation is subject to the same assumptions as for H1, including the linear dependence with having had fewer sons.